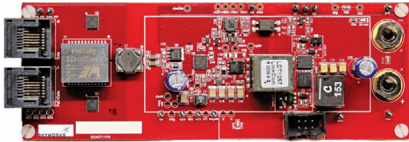


Nidhi Agarwal is a journalist at EFY. She is an Electronics and Communication Engineer with over five years of academic experience. Her expertise lies in working with development boards and IoT cloud. She enjoys writing as it enables her to share her knowledge and insights related to electronics with like-minded techies.

A Power Supply in a PoE



The evaluation board serves as a reference design for PoE applications, offering a 12V/6A output with 91% efficiency. It features integrated protection systems and supports up to 1000Mbps speed, optimising space and power management across conditions.

The Si34071 isolated forward topology evaluation board by Skyworks serves as a reference design for a power over Ethernet (PoE) powered device (PD) application, featuring the Si34071-FW-12V-C8 model that delivers a 12V/6A isolated output. It achieves 91% efficiency and meets IEEE 802.3bt and EMI standards, and includes features such as Autoclass, integrated transient overvoltage protection, and thermal shutdown protection. The compact 5mm x 5mm 32-pin QFN package optimises space. The DC/DC converter on the board features reset winding forward mode for low-power scenarios and active clamp forward mode for high-power requirements, ensuring efficient power management across conditions. Although the board does not support Ethernet data transmission, it connects secondary-side data from the Ethernet transformer to another connector, supporting speeds up to 1000Mbps through the PoE+ Data RJ45 connector.

The reference design is suitable for a power supply within a PoE powered device (PD) application.

Skyworks

<https://www.electronicshobby.com/electronics-projects/reference-design-for-a-power-supply-in-a-poe>

PoE Plus Power Design



The design integrates power conversion and protection for telecom applications, supporting high-speed operations and synchronisation capabilities. It offers power management, achieving efficiency and output.

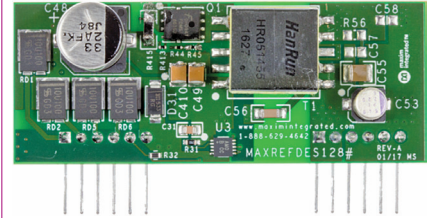
The ADM00844, a PoE plus powered device (PD) reference design by Microchip, simplifies the design process with elements for power conversion and protection for telecom applications. The design supports high-speed operation, controlling a sub-25ns pulse width on the gate output pin, operates with an internal oscillator over 2.5MHz, and achieves higher frequencies through synchronisation. It also minimises power dissipation during fault conditions with a fast 34ns response time from current sense to output and maintains efficiencies over 90% at high-output currents with a low quiescent current of 1.3mA. The design includes a soft-start function, a resistor-programmable current sense threshold, selectable soft-start retry, and a robust output driver. It performs front-end handshaking to accept power from the PSE when a powered Ethernet cable is connected, generating an 18.3V output with up to 1.7A of output current from input voltages between 36V and 57V.

The reference design is suitable for telecommunications, networking, security systems, wireless access, and industrial automation.

Microchip

<https://www.electronicshobby.com/electronics-projects/poe-plus-power-design-reference-design>

PoE Flyback Power Supply



The PoE module reference design delivers 5V at 12.5W from a 36V to 57V input, featuring a no-opto isolated flyback converter with 85% efficiency. Suitable for various stages of development, it offers control and power management capabilities.

The MAXREFDES128 PoE module reference design from Analog Devices outputs 5V at 2.5A (12.5W) from an input range of 36V to 57V, suitable for evaluation, design, prototyping, and production. It includes control circuitry and electrical components necessary for constructing an isolated flyback converter power supply. The module comprises two sections: the PD controller and the no-opto flyback DC-DC converter. The no-opto flyback DC-DC converter uses a 60V peak current mode, fixed-frequency switching controller designed for discontinuous conduction mode (DCM) and achieves 85% efficiency at a 100kHz switching frequency. It includes a primary-side MOSFET for energy storage in the transformer's winding and a secondary-side synchronous MOSFET driver for efficiency and thermal management. The design also features startup and input overvoltage protection settings with a 5ms soft start to reduce PSE inrush current, utilises a resistive divider for setting switching thresholds, and includes two 6-pin connectors and two bridge rectifiers for compatibility with various PSE types.

The reference design is suitable for stable power delivery across various applications.

Analog Devices (ADI)

<https://www.electronicshobby.com/electronics-projects/reference-design-for-poe-midspan-or-endpoint-insertion>